Sub-PRD 4 — Client Admin — Reporting & Analytics

> ⚠️ **V1 SCOPE LOCKED — March 20, 2026**
> **V1 reports: R3, R5, R6, R7 only.**
> **R1 (Overview Dashboard), R2 (Department Completion), R4 (Overdue Learners): V2** — depend on learning paths.
> All reports: Client Admin scope only. Team Manager: V2.

Document Type: PRD — Detailed Feature Spec with Acceptance Criteria

Parent Document: Client Admin Master PRD

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Status: PARTIAL IN SCOPE — 4 of 7 reports in V1

V1 Report Set (locked)

#	Report	V1	V2	Reason for deferral
R3	Per-Learner Score & Attempts	✅		Available from content_assignments + attempts
R5	Content Item Performance	✅		Available from attempts data
R6	Certificates Issued Log	✅		Available from certificates table
R7	Learner Activity Log	✅		Available from learners + login data
R1	Tenant Overview Dashboard		✅	Needs learning path enrollment data

R2	Department Completion Rates		✅	Needs learning path completion rates
R4	Overdue Learners		✅	Needs learning path deadlines

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1. Problem This PRD Solves

A Client Admin who has onboarded learners and assigned content needs to answer: **"Is learning actually happening — and where is it failing?"**

V1 answers the most immediately actionable questions: individual learner performance, content difficulty, certificate tracking, and learner engagement (activity).

V2 will add the org-level completion and overdue learner views that require learning path data.

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2. Locked Decisions (V1)

Decision	Value
V1 report set	R3, R5, R6, R7
Team Manager scoped reports	V2 — all V1 reports are Client Admin only
Data freshness	Near real-time (max 5 min cache lag)
Export format	CSV only
Scheduled reports	V2
Custom report builder	V2
Default date range	Last 30 days

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3. RBAC Scoping (V1)

All reports are **Client Admin only** in V1. Team Manager scoped reports: V2.

All report API endpoints inject `tenant_id` from middleware. No cross-tenant access possible.

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4. Report Specifications (V1)

R3 — Per-Learner Score & Attempts

Consumer: Client Admin

Entry: Reports → Learner Performance

Columns:

Column	Definition	Sortable
Learner Name	Full name (links to learner profile)	Yes
Department	Department name	Yes
Content Item	Assessment/course title	No
Type	Assessment / Course	No
Attempts	Total attempts	Yes
Best Score	Highest score %	Yes
Last Attempt	Date of most recent attempt	Yes
Status	Passed / Failed / In Progress / Not Started	Yes

One row per (learner × content item).

Filters: Date range, Department, Content type (Assessment/Course), Status, Learner search

Sort default: Last Attempt descending

Acceptance Criteria:

```
Given learner Priya attempted "NEET Biology Mock Test 1" 3 times:
    Attempt 1: 45% (Failed), Attempt 2: 58% (Failed), Attempt 3: 72% (Passed)
When I view R3
Then one row shows: Attempts=3, Best Score=72%, Status=Passed
```

R5 — Content Item Performance

Consumer: Client Admin

Entry: Reports → Content Performance

This is the key differentiator metric — surfaces which content is poorly aligned to learner level.

Columns:

Column	Definition	Sortable
Content Title	Item title	Yes
Type	Course / Assessment	No
Category	Exam category	Yes
Total Attempts	All attempts across all learners	Yes
Unique Learners	Distinct learners who attempted	Yes
Avg Score	Mean of all attempt scores	Yes
Pass Rate %	(Passers / Unique learners) × 100	Yes
Avg Attempts to Pass	Mean attempts before first pass (passers only)	Yes

Filters: Date range, Category, Type, Pass rate range

Sort default: Pass Rate % ascending (worst performers first — most actionable)

Acceptance Criteria:

```
Given "NEET Physics Mock Test 1" attempted by 10 unique learners, 7 passed
Then Pass Rate % = 70%
```

R6 — Certificates Issued Log

Consumer: Client Admin
Entry: Reports → Certificates

Columns:

Column	Definition	Sortable
Learner Name	Full name	Yes
Department	Department name	Yes

Content Title	Course title	Yes
Certificate Number	KSS-{tenant}-{YYYYMMDD}-{seq}	No
Issued Date	certificates.issued_at	Yes

V1 note: Certificates issued for course completion only. Assessment + path certificates: V2.

Filters: Date range, Department, Learner search

Sort default: Issued Date descending

Acceptance Criteria:

```
Given Priya completed "HIPAA Compliance Training" on March 20, 2026
Then one row appears: Learner=Priya, Course=HIPAA..., Cert No=KSS-AKS-20260320-001, Date=Mar 20
```

R7 — Learner Activity Log

Consumer: Client Admin
Entry: Reports → Learner Activity

Columns:

Column	Definition	Sortable
Learner Name	Full name	Yes
Department	Department name	Yes
Team	Team name	No
Status	Active / Inactive	Yes
Last Login	Most recent login timestamp	Yes
Last Active	Most recent platform action	Yes
Days Since Active	TODAY − last_active_at	Yes

"Never active" learners: last_active_at = NULL. Shown at top when sorted by dormancy (NULL = most dormant).

Filters: Department, Status (Active/Inactive/All), Days since active range, Learner search

Sort default: Days Since Active descending (most dormant first)

Acceptance Criteria:

```
Given learner Ananya was added 15 days ago and never logged in
And Rohit last logged in 8 days ago
When sorted by Days Since Active descending
Then Ananya appears above Rohit
And Ananya's Days Since Active = "Never"
```

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5. V2 Reports (Deferred — DO NOT BUILD IN V1)

R1 — Tenant Overview Dashboard (V2)

Depends on: learning path enrollment counts, completion rates, overdue learner counts.
Deferred because learning paths are V2.

R2 — Department Completion Rates (V2)

Depends on: learning path enrollment + completion per department.
Deferred because learning paths are V2.

R4 — Overdue Learners (V2)

Depends on: learning path deadline_date + enrollment status.
Deferred because learning paths + deadlines are V2.

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6. CSV Export Spec (V1)

Every V1 report has an "Export CSV" button.
Export applies all currently active filters.
Sync for ≤10,000 rows. Async (email notification) for >10,000 rows.
Filename: `keyskillset_{report_slug}_{YYYYMMDD_HHMMSS}.csv`

Report	Slug	Key Columns
R3	learner_scores	learner_name, email, department, content_title, content_type, attempts, best_score_pct, last_attempt_date, status
R5	content_performance	content_title, content_type, category, total_attempts, unique_learners,

		avg_score_pct, pass_rate_pct, avg_attempts_to_pass
R6	certificates	learner_name, email, department, content_title, certificate_number, issued_date
R7	learner_activity	learner_name, email, department, team, status, last_login_date, last_active_date, days_since_active

CSV format: UTF-8 with BOM, comma delimiter, ISO 8601 dates, numeric percentages (no % symbol).

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7. Report Navigation (V1)

```
Reports (sidebar)
├── Learner Performance    ← R3
├── Content Performance    ← R5
├── Certificates           ← R6
└── Learner Activity       ← R7
```

V2 additions: Overview Dashboard (R1), Department Completion (R2), Overdue Learners (R4).

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8. Audit Events

Action	Trigger
REPORT_EXPORTED	CSV export downloaded

End of PRD 4 — Reporting & Analytics v2.0 (V1 scope locked March 20, 2026)
V1: R3, R5, R6, R7 only | R1, R2, R4: V2 (need learning paths) | Team Manager scoped reports: V2